

Qatar-1b

R-2810

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_120809 · created 2026-09-05T01:16:52+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456148.90184 +/- 0.00016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.055
Transit depth (Rp/Rs)^2	2.72 +/- 1.81 [%]
Orbital Inclination (inc)	83.25 +/- 2.89
Airmass coefficient 1 (a1)	3.91 +/- 1.78
Airmass coefficient 2 (a2)	-1.04 +/- 0.34
Scatter in the residuals of the lightcurve fit is	34.65 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	6.75
Transit Duration (day)	0.059 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.055 vs 0.1463 (deviation 0.28σ)
Duration fit vs published	0.059 d vs 0.0692 d (deviation 0.35σ)
Mid-time O–C	0.3 min
Depth SNR	2.5
Residual scatter	34.65 %

Light curve

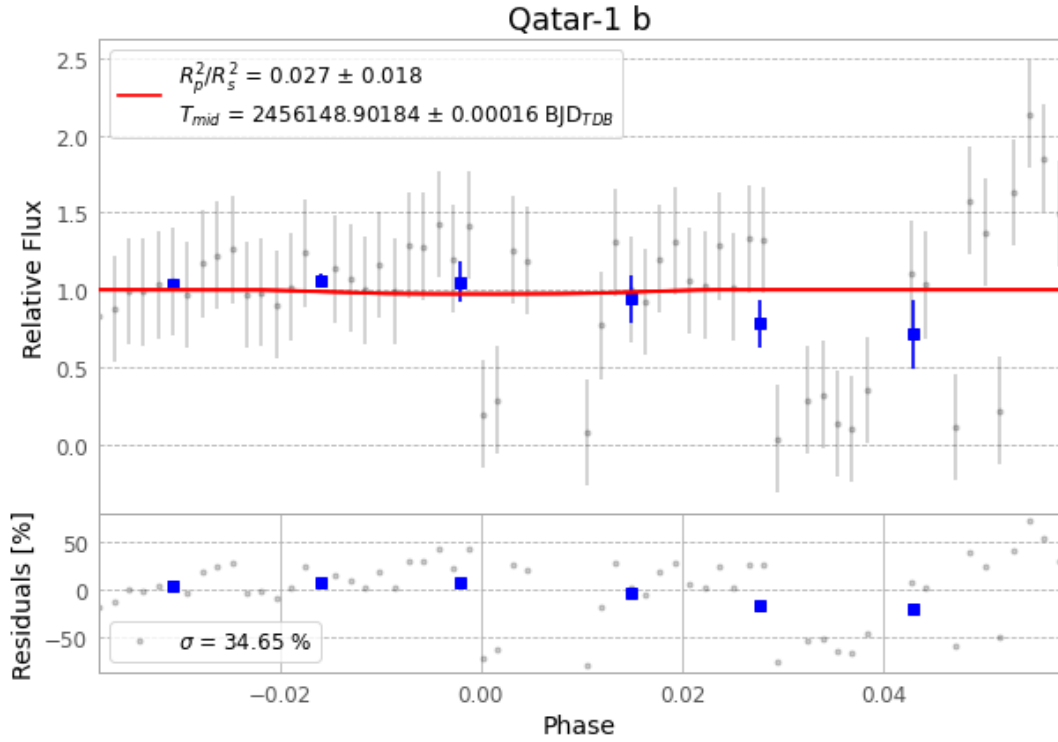


Figure 1 — FinalLightCurve_Qatar-1 b_2012-08-09.png (SHA-256 a7c5a08cdca4d62e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [539, 292], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6aeacf64dd4dccb9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), Qatar-1120809081812.FITS → Qatar-1120809113312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-120809.log (f70db2d2e5f8...), FinalLightCurve_Qatar-1 b_2012-08-09.png (a7c5a08cdca4...), FinalLightCurve_Qatar-1 b_2012-08-09.csv (1baf2c3c626c...)

Compute resources

Wall time 507.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 01:16Z.